6 Months Training TR-104 Full Stack Development

January, 2026

Day 16 – Firebase Authentication (Sign-In, Sign-Up, and Google Auth)

1. Objective of the Day

The objective was to implement complete authentication functionality using Firebase, including user sign-in, sign-up, and third-party authentication.

2. Tasks Performed

- Created Sign-In component using Reactive Forms
- Implemented email/password login using Firebase authentication
- Created Sign-Up component for new user registration
- Registered users using Firebase createUserWithEmailAndPassword
- Redirected users after successful login and registration
- Implemented Google authentication using Firebase provider
- Enabled users to sign in using Google accounts
- Managed authentication state during form submission
- Displayed loading indicators and disabled buttons during requests

3. Tools / Technologies Used

- Angular Framework
- Firebase Authentication
- AngularFire
- Reactive Forms

- TypeScript

4. Key Learnings

- Firebase provides secure authentication services
- Reactive Forms integrate well with authentication workflows
- Third-party authentication improves user experience
- Managing UI state during authentication enhances usability
- Proper handling of authentication flow is essential

5. Challenges Faced

- Handling authentication flow correctly
- Managing asynchronous operations during login and registration

6. Solutions

- Followed structured implementation for authentication
- Tested each authentication method separately

7. Conclusion

- Successfully implemented Firebase authentication system
- Enabled both email/password and Google login functionality